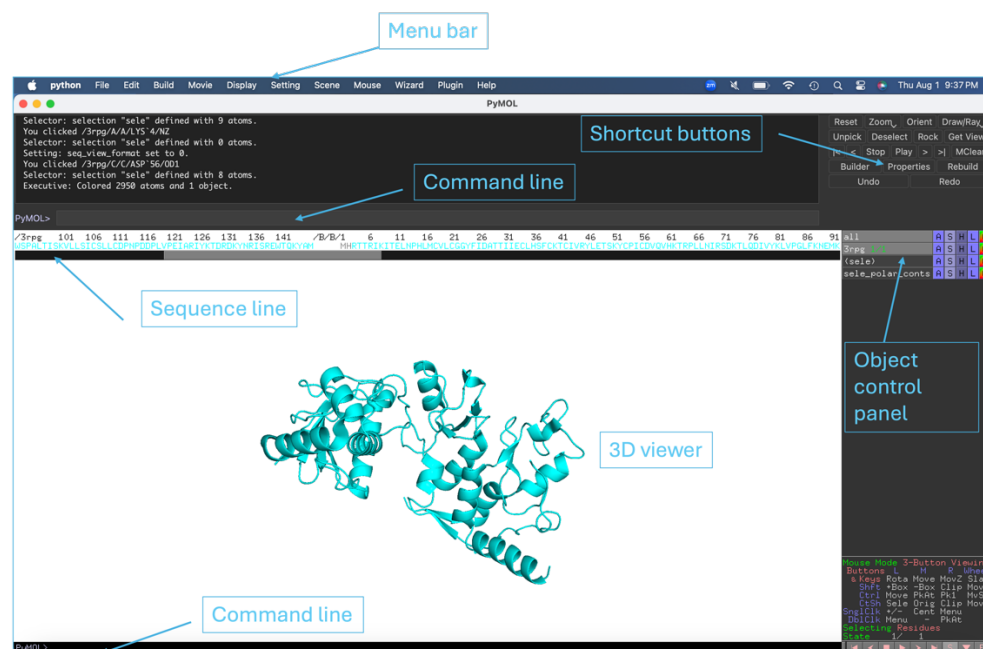


PyMOL exercise

You will look at the 3D structures of three interacting proteins including RING2 (chain C) and BMI1 (chain B), both being part of human polycomb repressive complex 1-like, which plays a role in histone and gene regulation as describe by Wang et al. (2004) in “Role of histone H2A ubiquitination in polycomb silencing”. These two proteins interact as a heterodimer with the E2 enzyme Ubch5c (chain A) and the complex structure has been crystallized by Bentley et al. (2011) “Recognition of Ubch5c and the nucleosome by the BMI1/RING1b ubiquitin ligase complex” (PDB ID 3RPG).

Now you will analyze how these proteins interact with each other.

The PyMOL Window has the following parts:



- **Menu bar** – Pull down menus
- **Command line** – written commands and scripting for experienced users
- **Sequence line** – if a sequence is displayed as Residue codes, it shows 1-letter code
- **Viewer** - molecular graphic window, where the molecule is visualized
- **Shortcut buttons** – common commands (e.g., Draw/Ray)
- **Object Control Panel** – Object name and 5 Object buttons
- Each object can be controlled using the **A**(action), **S**(show), **H**(hide), **L**(label) and **C**(color).

Open the Protein Data Bank (PDB) <https://www.rcsb.org/> webpage.

Write **3RPG** in the search field and perform the search to find the structure with PDB ID 3RPG. Click on the **3RPG** -link and select **Download Files – Legacy PDB Format** to save the structure on your computer in an appropriate folder.

Open **PyMOL**

Open the structure in PyMOL: In the Menu bar, select **File/Open/3RPG.pdb** from the directory where you save the file.

Zoom and center, and tips:

At all times, to **unselect everything**, click on the background of the 3D viewer.

In PyMOL you can move around to visualize the protein. In these instructions we use the following abbreviations:

- Left Mouse Button = LMB
- Middle Mouse Button (mouse wheel) = MMB
- Right Mouse Button = RMB

You can move the molecule

Hold on the LMB and drag the cursor in the window

Now you can zoom in

Hold on the RMB and drag the cursor in the window

You can adjust z-direction

Scroll (Careful, not too fast!) with the mouse wheel

You can center your molecule, center rotation on molecule center, automatically zoom

3RPG → select A/center

3RPG → select A/zoom

Change background color

Menu bar **Display/Background/White**

Tips

To color, be careful to use contrast colors for easier visualization, and do not use **blue** (nitrogen color) or **red** (oxygen color).

When creating a picture, less is better, keep visible only what is necessary for your picture. E.g. if hydrogens are included in the structure, it is good to hide them during visualization.

Note: 3RPG does not have hydrogens. White background is used in articles and transparent background is good for presentations.

The PyMOL Viewer automatically shows the secondary structure of protein as one-color cartoon and all the water as red crosses, which is not very informative in a large molecule.

Change the protein view:

Change representation:

3RPG → select H/everything (*Start with an empty 3D Viewer*)

3RPG → select S/cartoon

Under **S(Show)**, you can find ribbon and cartoon styles for secondary structure, and several atom styles e.g., lines and sticks.

Color by chain:

3RPG → select C/by chain/**by chain** (elem C)

You can also color the whole protein complex in one color or only select some residues to color those in another color.

Find interactions between chain:

Display sequence

On Menu bar, Display/Sequence

On Menu bar, Display/Sequence Mode/Chain Identifiers

What happens?

Find interactions

Select chain C

New object (**sele**) is created

What else happens?

(sele) → A/find/polar contacts/to other atoms in object

New object created sele_polar_conts

sele_polar_conts → A/rename

The prompt appears on the upper left in your viewer window. Remove “**sele_polar_conts**” with backspace, write **Contacts-ChainC** and **<Enter>**

What happens?

To change color for interactions: **Contacts-ChainC** → C/grays and select black

You can search protein-protein interactions, DNA-protein interactions, protein-ligand interactions in a similar way. You can find polar interactions within chain, with ligands, water, etc.

To visualize interactions:

3RPG → S/lines

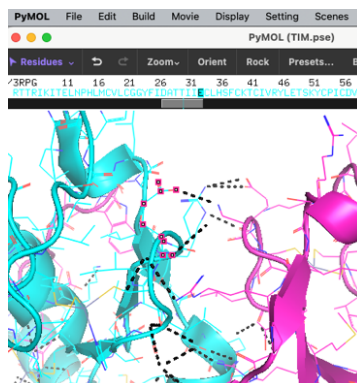
Label important residues:

Unselect everything by clicking on the background of the 3D viewer

On Menu bar Display/Sequence Mode/Residue codes

Select **Glu33 in Chain B** and **Lys65 in Chain C** by scrolling the Sequence line

Pymol figure shows
how Glu33 in Chain B is selected
in the Sequence line and in the 3D View



New object (**sele**) is created

(**sele**) → L/Label/residues

(**sele**) → S/sticks

(**sele**) → A/zoom

What happens?

Unselect everything, click on the background of the 3D viewer

You can select residues by clicking them in the 3D viewer or in the Sequence line.

Save:

Save a session:

It is recommended to save session regularly, since PyMOL can crash.

File/Save Session As ...

Select the right folder and give a name e.g., pymol_exercise

High quality pictures:

Create a high-quality picture with the Glu33-Lys65 interaction:

Hide all other residues than Glu33 in Chain B and Lys65 in Chain C

3RPG → H/lines

Glu33 in Chain B interacts with a water, which is not visible. Show waters

3RPG → S/nonbonded

To show the water as a sphere, select the water on the 3D viewer (the red cross connected with black dashed line to Glu33) and

(sele) → S/nb_spheres

3RPG → H/nonbonded

Unselect everything, click on the background of the 3D viewer

Move around to create a picture of labelled interaction.

Choose the **Draw/Ray button** on Shortcuts button panel.

Click **Reset** to set size of picture to your Pymol window size

Change 150 DPI to **300 DPI**

Click on **Ray (slow)**

Save image to a file: give a name and save in an appropriate folder.

Scenes:

Save a scene:

On Menu bar, select **Scene/Store/F1**

You can **Recall** the scene later by clicking **F1 on the keyboard** or **the 3D viewer**

Save the session.

You have learned the basics of PyMol now and it is time to have a Group meeting and start working on the project work!